

TECHNICAL SKILLS

Languages: Java, Python, SQL, C#
Frameworks: Spring Boot, FastAPI, React, LangChain
Backend: REST APIs, Hibernate, JPA
Databases: PostgreSQL, MySQL, MongoDB
Cloud & Infrastructure: Azure, Docker, Linux
Tools: Git, IntelliJ, Maven, CI/CD
Concepts: System Design, Distributed Systems, Microservices Architecture, Data Structures & Algorithms, Dependency Injection

EDUCATION

Bauhaus-Universität Weimar
Master of Science in Digital Engineering (Grade: 1.6)

- Relevant Coursework: Machine Learning, Computer Vision, Simulation Methods, Stochastic Systems, Optimization, Data Structures & Algorithms, Object-Oriented Programming, Software Engineering

Germany
04/2023 – 09/2026

Anna University Chennai
Bachelor of Engineering in Civil Engineering

- Co-author, *Int. J. Steel Struct.* (2024), DOI: 10.1007/s13296-024-00816-9

India
04/2017 – 03/2021

PROFESSIONAL EXPERIENCE

FORVIA Hella GmbH & Co. KGaA
Software Developer Intern / Master's Thesis

- Designed and built a distributed Python workflow engine to orchestrate automotive simulation jobs across HPC clusters
- Developed modular backend components for job scheduling, execution monitoring, and result aggregation
- Automated end-to-end simulation pipelines, eliminating 90% of manual processes and reducing overhead
- Integrated external APIs and data pipelines to connect simulation infrastructure with downstream systems
- Refactored legacy simulation scripts into structured backend services, improving scalability and reducing technical debt

10/2024 – 09/2025
Lippstadt, Germany

Bauhaus-Universität Weimar
Student Assistant

- Enhanced a Retrieval-Augmented Generation (RAG) system by integrating external knowledge sources and improving retrieval quality
- Supported evaluation and teaching activities at the Chair of Stochastics and Optimization

04/2024 – 09/2024
Weimar, Germany

Tata Consultancy Services
Assistant System Engineer

- Developed and maintained Java Spring Boot microservices for a US-based financial services client, supporting high-volume transaction processing across distributed systems
- Implemented automated reconciliation pipelines to validate financial transactions against external records, identifying discrepancies and reducing manual exception handling
- Built backend APIs and batch processing workflows to handle large-scale financial data, ensuring accuracy and reliability across end-of-day settlement cycles

04/2021 – 02/2023
Chennai, India

PROJECTS

RAG Chatbot Backend Application | *LangChain, FastAPI, Python, ChromaDB, LLM*

- Built a Retrieval-Augmented Generation (RAG) system enabling efficient querying of large document datasets
- Implemented backend APIs for document ingestion, vector indexing, and LLM-based response generation
- Designed conversational workflows with chat history storage using SQLite and support for follow-up queries

Order Management System | *Spring Boot, Java, PostgreSQL*

- Developed a backend system to manage users, products, and orders using REST APIs
- Designed relational data models with entities such as Order, Product, and User using Spring Data JPA
- Implemented order workflows including item management, total price calculation, and status tracking

COMPETITIONS

- Participated in TREC 2024 RAG Track, organized by National Institute of Standards and Technology (NIST), USA and University of Waterloo

LANGUAGES

German (B1/B2), English (C1), Tamil (Native)